

Course Code : CST 364

QTSR/RW – 24 / 1666

**Sixth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

OBJECT ORIENTED DATA STRUCTURE

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated.
- (3) Assume suitable data wherever necessary.

1. (a) Determine the output of below Java code, if it is written correctly or find the error, if any :

```
(i) public class Test
    {
        double i, j ;
        public static void main(String[ ] args) {
            Test object = new Test( ) ;
            object.j = ++ object.i + object.i + + - + + object.j ;
            System.out.println(object.i);
            System.out.println(object.j);
        }
    }

(ii) public class Main
    {
        public static void main (String[ ] args)
        {
            int x = 4, y = 0, z ;
            while (x >= 0)
            {
                if (x == y)
                    break ;
                else
                    System.out.println(x + " " + y);
                x -- ;
                y ++ ;
            }
        }
    }
```

1+1(CO1)

- (b) State the different level of access protection available in Java. 2(CO1)
- (c) Design a Package myPackage1 with class Employee including attributes employee_name, employee_Id and another package myPackage2 with interface incentive. Write a Java program with subclass name Manager which extends "Employee" class and implement "incentive" interface from other package. Define method calculateSalary() to find the total salary of manager. Create a demo class with main() method and demonstrate the use of all functions. 3(CO1)
- (d) Explain any Four methods of String class with syntax. Also write a Java program to reverse a string given by user as input. 3(CO1)
2. (a) Create a class Student with protected data members as Students_name and Student_ID, marks_Subject1, marks_Subject2 and marks_Subject3 as a data members. Create a constructor and method called findPercentage based on marks of 3 subjects to calculate the percentage of student. Create a demo class with main() and demonstrate the use of all functions and display percentage for 3 Engineer Students. 4(CO2)
- (b) Create a class called Product having attributes such as product_id, product_name and cost. Include following :
- (I) Constructor.
 - (II) A method set_data() to input the above information of product.
 - (III) A method display() to print the information of entered values.
 - (IV) A method find_cost() to display the information of product having a cost greater than 1000 Rs.
- Write a complete Java program to test the class using array of object for 10 product. 4(CO2)
- (c) Explain method overloading in Java with a suitable example program. 2(CO2)
3. (a) Explain the term Polymorphism in Java. Give the difference between compile time and run time Polymorphism. 2(CO2)

- (b) Write a Java program to create an interface Resizable with methods `resizeWidth(int width)` and `resizeHeight(int height)` that allow an object to be resized. Create a class Rectangle that implements the Resizable interface and implements the resize methods. 4(CO2)
- (c) Write a Java program to create an abstract class RentVehicle with abstract methods `rent()`, `return()` and `calculateRentalCost()`. Create subclasses Car and bike that extend the RentVehicle class and implement the respective methods to calculate RentalCost and display information for each vehicle. 4(CO2)
4. (a) Explain the concept of asymptotic notation in the context of algorithm analysis. Provide examples to illustrate the difference between the following asymptotic notations : O-notation (Big O), Ω -notation (Big Omega) and Θ -notation (Big Theta). 3(CO3)
- (b) Each student in a class of 30 student takes 6 tests in which scores range between 0 to 100. Suppose the test scores are stored in a 30x6 array Test. Write a module which –
- (I) Finds the average grade for each student.
 - (II) Finds the final grade for each student where the final grade is the average of the student's five highest test score.
 - (III) Finds the number NUM of students who have failed, i.e. whose final grade is less than 60.
 - (IV) Finds the average of the final grades. 3(CO3)
- (c) Find the complexity for the given code and elaborate :
- ```

(i) for (i = 1 ; i <= n ; i++)
 for (j = 1 ; j <= n ; j++)
 {
 C[i][j] = 0 ;
 for (k = 1 ; k <= n ; k++)
 C[i][j] = C[i][j] + a[i][k]
 *b[k][j]
 }

```

```
(ii) avg(a, n) {
 Sum = 0 ;
 for (i = 1 ; i <= n ; i ++)
 read (a[i]) ;
 for (i = 1 ; i <= n ; i ++)
 Sum = Sum + a[i] ;
 avg = Sum / n ;
 print(avg) ; }
```

4(CO3)

5. (a) Convert Infix expression  $((A+B)*D+E/(F+G*D))+C$  to its Postfix expression using stack. Show all the steps of conversion and evaluation using stack data structure. Also write an algorithm for the same. 5(CO4)
- (b) What is difference between Circular queue and Simple queue ? Write an algorithm or Java pseudo code for deleting an element from circular queue. 5(CO4)
6. (a) What is the difference between singly linked list and doubly linked list ? Write an algorithm for inserting the node in doubly linked list. 5(CO4)
- (b) Write an algorithm or Java function to count the number of nodes in singly circular linked list. 5(CO4)

